

Muhammad Rayyan

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Lahore, Punjab, Pakistan

Education & Certificates

- Bachelor's Degree in Data Science, Khwaja Fareed University of Engineering and Information Technology (KFUEIT) Graduation – 2026
- Certificate: Python Programming for All Levels
- Certificate: Google AI Essentials
- Certificates: Data Science Methodology, Tools for Data Science, What is Data Science?

Work History

Full Stack Developer at Journal Post Group, Lahore

July 2026 to Current

- Building JournalPost.co, a viral media platform, using Next.js for the frontend and Payload CMS for content management to keep editorial teams moving quickly.
- Designed the data layer with Supabase PostgreSQL to support fast-loading, highly shareable content built for scale.
- Structured the CMS and frontend architecture to prioritize speed and shareability, aligning technical decisions with content virality goals.

AI/ML Intern at Voliom, Rawalpindi

June 2025 to September 2025

- Developed a computer vision platform backend API using FastAPI and PostgreSQL, working closely with a senior developer on production code.
- Built a complete authentication system with Firebase integration, email-based user verification, and JWT-based security to protect user data.
- Implemented project management features enabling users to create and manage computer vision datasets and upload images via Firebase Storage.
- Integrated Google Vertex AI to power image generation from text prompts and image-to-text analysis capabilities.
- Modeled the database using SQLAlchemy and containerized the application with Docker for consistent, portable deployments.
- Applied RESTful API design principles and strengthened backend development skills through mentorship and structured code reviews.

Projects

Imagenix – Final Year Project

- Architected a three-service monorepo integrating Stable Diffusion, ControlNet, and Grounding DINO for AI-powered image generation.
- Used BullMQ and Redis to manage asynchronous job queues across services, keeping long-running AI tasks responsive.
- Implemented S3-compatible storage to handle generated image assets across the distributed system.

Universal Flexible Backend

- Built a schema-less document storage backend with FastAPI and MongoDB, allowing dynamic data models without rigid schema constraints.
- Deployed the service on Railway to provide a flexible, reusable backend foundation for multiple future projects.

Weather App (mrayyan911/weather-app)

- Built a Python weather application using FastAPI, SQLite, and APScheduler for scheduled data updates.
- Migrated the weather data source to Open-Meteo after a third-party API outage, restoring service reliability.
- Wrote automated tests using pytest fixtures and mocking patterns, and deployed the application on Railway.

Skills

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS

Frontend: React, Next.js, Redux, React Router, Tailwind CSS, Three.js

Backend: Node.js, NestJS, Django, FastAPI

Databases & Storage: PostgreSQL, MongoDB, Redis, MinIO

Tools & DevOps: Docker, Git, GitHub, CI/CD, Prisma, SQLAlchemy, BullMQ, Turborepo

Deployment: Render, Railway, Vercel